



Hemoglobin amino acids by the numbers

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EDITORIAL

Hemoglobin amino acids by the numbers

There exist several numbering systems for nucleotides in DNA and for amino acids in proteins [1]. For DNA, there are several levels of variation; where to start counting, how to handle introns and different reference sequences to mention a few. This diversity, of course, has historical reasons [2], but it complicates literature searches for specific sequence variants found in clinical laboratory routine or research, as such sequence variants may be denoted in several different ways in publications and databases. At the protein level, sources of numbering confusion are fewer than at the DNA level, but amino acid numbering is still not uniform.

In this issue of The Scandinavian Journal of Clinical & Laboratory Investigation, Lenters-Westra et al. [3] nicely investigate HbA1c measurements by several methods in the presence of the β -globin variant Hb Tacoma. For hemoglobin, the most commonly used amino acid numbering system is the Common protein-based description [1], which denotes the first amino acid *after* the initiator methionine as number 1. With this numbering system β -globin giving rise to HbS, the sickle cell disease hemoglobin, is denoted *beta 6 Glu > Val*, i.e. glutamic acid in position 6 is changed to valine, and β -globin of Hb Tacoma is designated *beta 30 Arg > Ser*. This numbering system is for instance used by the valuable HbVar database [4] frequently consulted during hemoglobinopathy evaluation.

A reason for using the Common protein-based description is that the initiator methionine of the β -globin chain (and the initiator methionine of the α -globin) is removed during protein synthesis [5] rendering valine as the amino terminal residue in the mature sequence and consequently number 1.

In the Official HGVS protein-based description, however, the initiator methionine is represented as number 1 [6]. Counting is therefore shifted one step forward relative to the Common protein-based description, leaving the sickle

mutation position as number 7 and Hb Tacoma to be altered in amino acid 31. The Official HGVS protein-based description has the advantage that amino acid numbering is in pace with cDNA based numbering, both starting with the initiation codon.

Lenters-Westra et al. [3] have chosen to use the Official HGVS numbering system in their article, despite this is not in line with tradition for hemoglobin. However, this is today actually the recommended amino acid numbering system and must be approved of.

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